



Name: _____

Teacher: _____

Class: _____

2015

HIGHER SCHOOL CERTIFICATE PRELIMINARY COURSE

ASSESSMENT TASK 3 – PRELIMINARY FINAL EXAM

Mathematics Extension 1

Time allowed: 2 Hours

(plus 5 minutes reading time)

Syllabus Outcomes	Assessment Area Description and Marking Guidelines	Questions
P3 PE3	chooses and applies appropriate arithmetic, algebraic, graphical , trigonometric and geometric techniques solves problems involving polynomials	1, 2
P4	chooses and applies appropriate arithmetic, algebraic, graphical and trigonometric techniques	3
P6 P7	relates the derivative of a function to the slope of its graph determines the derivative of a function through routine application of the rules of differentiation	4
PE6	makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations	5

Total Marks 75

Attempt Questions 1-5

Question	Out of	Marks
Q1	/15	
Q2	/15	
Q3	/15	
Q4	/15	
Q5	/15	
	Percent:	

General Instructions:

- Questions 1-5 are to be started in a new booklet
- The marks allocated for each question are indicated
- In Questions 1-5, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used

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Question One (Start a new booklet)

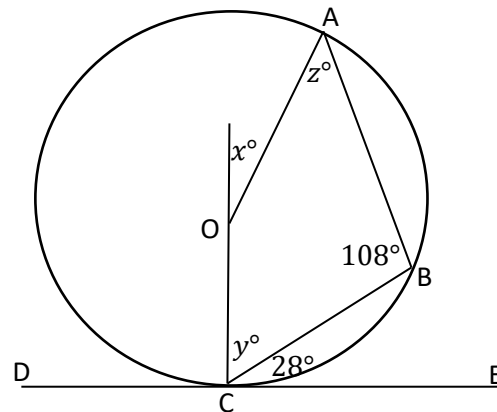
marks

(a) Solve

$$\frac{1}{x-2} > 1$$

3

(b)



O is the centre of the circle.

Find the value of each pronumeral giving reasons.

3

(c) Find the co-ordinates of the point that divides externally the interval joining the points $(-5, 6)$ and $(-2, 3)$ in the ratio $3:2$.

3

(d) The acute angle between the lines $2x + 2y = 5$ and $y = 3x + 1$ is θ . Find the value of θ , correct to the nearest minute.

3

(e) Let $P(x) = x^3 - ax^2 + x$ be a polynomial where a is a real number. When $P(x)$ is divided by $(x - 3)$ the remainder is 12. Find the remainder when $P(x)$ is divided by $(x + 1)$.

3

Question Two (Start a new booklet)

marks

- (a) Evaluate $\tan 225^\circ$. **1**
- (b) Simplify $\frac{1}{1-\sin V} + \frac{1}{1+\sin V}$ **3**
- (c) The bearing of a ship from lighthouse A is $N75^\circ E$. The bearing from a second lighthouse B, 44 km south of A, is $N40^\circ E$. Find the distance of the ship from B to the nearest km. You must draw a neat diagram marking all given information. **3**
- (d) A, B and C are three towns such that B is 20 km from A in a direction 330° and C is 30 km from A in a direction $203^\circ 08'$. Find the distance from B to C to one decimal place. You must draw a neat diagram marking all given information. **3**
- (e) The sides of a triangular field have lengths 80m, 90m and 100m.
- (i) Calculate the size of the smallest angle to the nearest minute. You must draw a neat diagram marking all given information. **3**
- (ii) Calculate the area of the field to the nearest m^2 . **2**

Question Three (Start a new booklet)

marks

(a) Write down the expansion for $\tan(\theta + \phi)$. **1**

(b) Simplify $\frac{1}{2} \sin 2\theta \tan \theta$ **2**

(c) If α and β are angles between 0° and 90° , and $\sin \alpha = \frac{3}{5}$ and $\tan \beta = \frac{7}{24}$, find without the use a calculator the exact value of $\sin(\alpha - \beta)$. **3**

(d) Solve for $0 \leq \theta \leq 2\pi$:

(i) $\sin 2\theta = \frac{-1}{2}$ **2**

(ii) $\sin^2 x - \sin x \cos x = 0$ **3**

(e)

(i) Write $\sqrt{3} \cos x - \sin x$ in form $r \cos(x + \alpha)$ where $0 < \alpha < \frac{\pi}{2}$. **2**

(ii) Hence or otherwise, solve $\sqrt{3} \cos x = 1 + \sin x$ for $0 < x < 2\pi$. **2**

Question Four (Start a new booklet)

marks

(a) Use the rules of differentiation to differentiate with respect to x :

(i) $y = (3x + 4)(x^2 - 2x)$ 2

(ii) $y = \sqrt{25 - x^2}$ (Answer as a surd) 2

(iii) $y = \frac{x+1}{x^2+1}$ 3

(b) Differentiate $f(x) = (x + 1)^3$ from first principles. 3

(c) A block is cast into the shape of a right cylinder with a total surface area of $20\pi \text{ cm}^2$. The radius of the base is r cm and the height is h cm. The total surface area of a cylinder is given by $A = 2\pi r^2 + 2\pi rh$.

(i) Express h in terms of r . 1(ii) Express the volume V in terms of r . 1(iii) Find the value of r for which the volume is the greatest. Answer in exact form. 3

Question Five (Start a new booklet)

marks

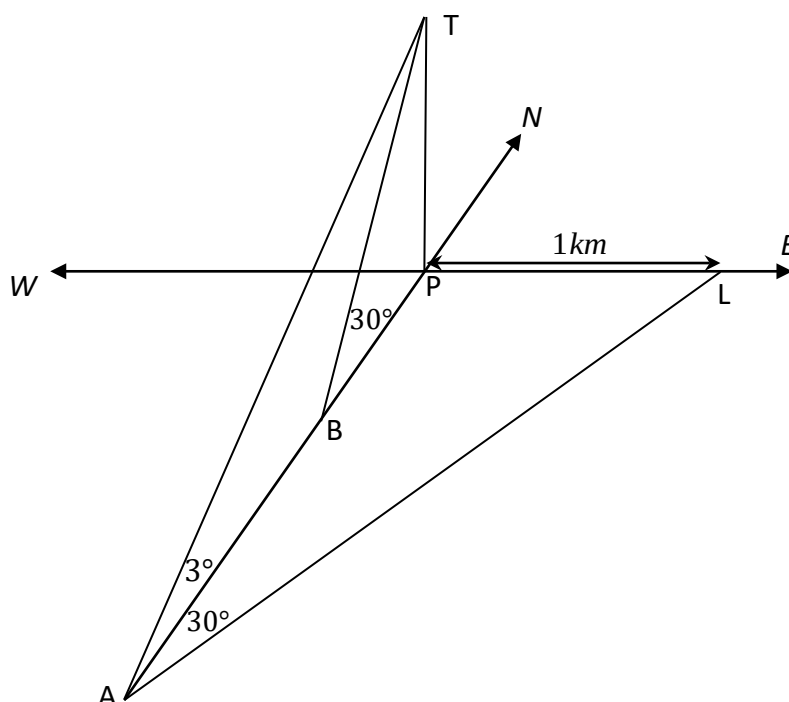
(a) Find the general solution to $\cos 2\theta = 2 + \cos \theta$.

3

(b) A boat is sailing due north from a point A towards a point P on the shore line. The shore line runs from west to east.

In the diagram, T represents a tree on a cliff vertically above P, and L represents a landmark on the shore. The distance PL is 1 km.

From A the point L is on a bearing of 030° , and the angle of elevation to T is 3° . After sailing for some time the boat reaches a point B, from which the angle of elevation to T is 30° .



(i) Show that $BP = \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}$

3

(ii) Find the exact distance AB.

1

(c) If $t = \tan\left(\frac{\theta}{2}\right)$ solve for t the equation $12 \tan \theta = 5$ for $\pi \leq \theta < \frac{3\pi}{2}$

4

(d) Sketch $y = x + \frac{4}{x-1}$, showing asymptotes, intercepts and stationary points having determined their nature.

4

Question One (Start a new booklet)

marks

(a) Solve $\frac{1}{x-2} > 1$

$$(x-2)^2 \times \frac{1}{x-2} > 1 \times (x-2)^2$$

$$x-2 > (x-2)^2$$

$$0 > (x-2)^2 - (x-2)$$

$$> (x-2)(x-2-1)$$

$$> (x-2)(x-3)$$

and graphing $y = (x-2)(x-3)$

Marking**Comments****3**

❶ correct algebraic processes to quadratic

Mostly well done.

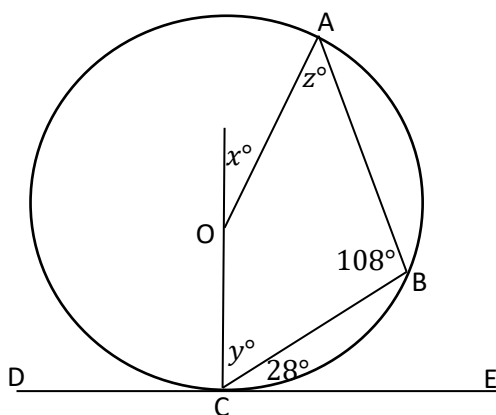
❶ Justifies (diagram or tests)

Some students did not justify their inequality

Then from graph, $(x-2)(x-3) < 0$ when $2 < x < 3$.

❶ solution

(b) O is the centre of the circle. Find the value of each pronumeral giving reasons.

3

$$x + 180^\circ = 2 \times 108^\circ \text{ (}\angle \text{ at cent. twice } \angle \text{ at circum.)}$$

$$x = 36^\circ$$

On same arc – reflex arc AC)

$$y + 28^\circ = 90^\circ \text{ (tang. and radius are } \perp \text{)}$$

$$y = 62^\circ$$

$$z + 144^\circ + 62^\circ + 108^\circ = 360^\circ \text{ (}\angle \text{ sum of quad.)}$$

$$z = 46^\circ$$

❶ correct with reason

❶ correct with reason

❶ correct with reason

If constructing or adding extra information into a given diagram, you MUST reproduce the diagram with the constructions or additions.

Many gave $z = 28^\circ$ and stated the Angle in Alternate Segment Theorem – this does not apply here!

$$\angle CAB = 28^\circ$$

- (c) Find the co-ordinates of the point that divides externally the interval joining the points $(-5, 6)$ and $(-2, 3)$ in the ratio $3:2$.

3

$$\begin{array}{c} (? , ?) \text{-----} (-5, 6) \text{-----} (-2, 3) \text{-----} (? , ?) \\ \text{-----} 3 \quad \text{-----} 2 \quad \text{-----} \end{array}$$

i.e. point is to the "right" of $(-2, 3)$

$$\begin{aligned} \text{hence } & \left(\frac{3 \times -2 + 2 \times -5}{3 + 2}, \frac{3 \times 3 + 2 \times 6}{3 + 2} \right) \\ & = (4, -3) \end{aligned}$$

① correct formula or other explanation
① -ve for external division correct
① solution

Many did not substitute into the formula correctly.

- (d) The acute angle between the lines $2x + 2y = 5$ and $y = 3x + 1$ is θ . Find the value of θ , correct to the nearest minute.

3

$$2x + 2y = 5 \quad \text{and} \quad y = 3x + 1$$

$$y = \frac{5}{2} - x \quad m_2 = 3$$

$$m_1 = -1$$

Hence

$$\begin{aligned} \tan \theta &= \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| \\ &= \left| \frac{3 - (-1)}{1 + 3 \times (-1)} \right| \\ &= \left| \frac{4}{-2} \right| \\ &= 2 \end{aligned}$$

$$\theta = \tan^{-1}(2)$$

$$= 63^\circ 26'$$

① correct gradients in formula

① correct tan value

① correct angle in degrees & minutes

Some students stated the formula incorrectly.

- (e) Let $P(x) = x^3 - ax^2 + x$ be a polynomial where a is a real number. When $P(x)$ is divided by $(x - 3)$ the remainder is 12. Find the remainder when $P(x)$ is divided by $(x + 1)$

3

$$P(x) = x^3 - ax^2 + x \text{ gives}$$

$$P(3) = 3^3 - a \times 3^2 + 3$$

$$12 = 30 - 9a$$

$$a = 2$$

Hence $x = -1$ gives

$$P(-1) = (-1)^3 - 2 \times (-1)^2 + (-1)$$

$$= -1 - 2 - 1$$

$$= -4$$

① $x = 3$ correctly substituted

① correct a value

① correct remainder

Mostly well done.

Question Two (Start a new booklet)

marks

(a) Evaluate $\tan 225^\circ$.

$$\tan 225^\circ$$

$$= +\tan 45^\circ$$

$$= 1$$

1

① correct value

Mostly well done

(b) Simplify $\frac{1}{1-\sin V} + \frac{1}{1+\sin V}$

$$\frac{1}{1-\sin V} + \frac{1}{1+\sin V}$$

$$= \frac{1+\sin V + 1-\sin V}{(1-\sin V)(1+\sin V)}$$

$$= \frac{2}{1-\sin^2 V}$$

$$= \frac{2}{\cos^2 V}$$

$$= 2\sec^2 V$$

3

① numerator correct

① resolves denominator

① answer

Occasional minor errors

A few could not resolve denominator into cos.

A non-fraction is simpler than a fraction! Many lost this last mark.

(c) The bearing of a ship from lighthouse A is $N75^\circ E$. The bearing from a second lighthouse B, 44 km south of A, is $N40^\circ E$. Find the distance of the ship from B to the nearest km. You must draw a neat diagram marking all given information.

3

44 km

$\angle SAB = 105^\circ$ (st. ln.), then $\angle ASB = 35^\circ$ (Δ sum), so

$$\frac{BS}{\sin 105^\circ} = \frac{44}{\sin 35^\circ}$$

$$BS = \frac{44 \sin 105^\circ}{\sin 35^\circ}$$

$$= 74.09777261$$

$$\doteq 74 \text{ km}$$

① diagram correct

① Sine Rule correct

① answer

Some students did not have North directly up the page for their bearing questions. Several also drew as if it was a 3D trig question.

Quite a few had issues putting A north of B in their diagram.

Many simple calculation errors.

READ the requirements of the question! Nearest km!

- (d) A , B and C are three towns such that B is 20 km from A in a direction 330° and C is 30 km from A in a direction $203^\circ 08'$. Find the distance from B to C to one decimal place. You must draw a neat diagram marking all given information.

3

 20 km 30 km

$$\angle BAC = 330^\circ - 203^\circ 08'$$

$$= 126^\circ 52'$$

$$BC^2 = 20^2 + 30^2 - 2 \times 20 \times 30 \times \cos(126^\circ 52')$$

$$= 1300 - 1200 \cos(126^\circ 52')$$

$$\doteq 2019.945863$$

$$BC \doteq 44.94380784$$

$$\doteq 44.9$$

❶ diagram
correct

❶ Cosine Rule
correct

❶ answer

Similar general
comments as for
(c) above.

READ – to 1 dp.
Note: Only one
mark deducted in
this question for
this type of error.

- (e) The sides of a triangular field have lengths 80m , 90m and 100m .

- (i) Calculate the size of the smallest angle to the nearest minute. You must draw a neat diagram marking all given information.

3

$$\cos \theta = \frac{100^2 + 90^2 - 80^2}{2 \times 90 \times 100}$$

$$= \frac{117}{180}$$

$$\theta = \cos^{-1}\left(\frac{117}{180}\right)$$

$$= 49^\circ 28'$$

❶ diagram
correct

❶ Cosine Rule
correct

❶ Answer to
nearest
minute

Many wasted
precious exam
time on this
question by not
realizing that the
smallest side is
opposite the
smallest angle!

Otherwise
generally well
done.

(ii) Calculate the area of the field to the nearest m^2 .

2

$$\begin{aligned}A &= \frac{1}{2} \times 100 \times 90 \times \sin(49^\circ 28') \\&= 3420.126015 \\&\doteq 3420 m^2\end{aligned}$$

① Area rule correct

① Answer

Generally well done,
although a few wanted to
use cos instead of sin.

Question Three (Start a new booklet)

marks

(a) Write down the expansion for $\tan(\theta + \phi)$.**1**

$$\tan(\theta + \phi) = \frac{\tan \theta + \tan \phi}{1 - \tan \theta \tan \phi}$$

1

Well done.

(b) Simplify $\frac{1}{2} \sin 2\theta \tan \theta$ **2**

$$\begin{aligned} & \frac{1}{2} \sin 2\theta \tan \theta \\ &= \frac{1}{2} \times 2 \sin \theta \cos \theta \times \frac{\sin \theta}{\cos \theta} \\ &= \sin^2 \theta \end{aligned}$$

1 correct substitutions
1 Answer

Generally well done.

(c) If α and β are angles between 0° and 90° , and $\sin \alpha = \frac{3}{5}$ and $\tan \beta = \frac{7}{24}$, find without the use a calculator the exact value of $\sin(\alpha - \beta)$.

3

Many did not draw diagrams!

Some wrote

$$\sin \frac{3}{5} \text{ instead of}$$

$$\sin \alpha = \frac{3}{5}$$

$$\sin \beta = \frac{7}{25}, \cos \beta = \frac{24}{25}$$

$$\sin \alpha = \frac{3}{5}, \cos \alpha = \frac{4}{5}$$

1 setup diagrams and values correct

$$\text{Then } \sin(\alpha - \beta) = \sin \alpha \cos \beta - \sin \beta \cos \alpha$$

$$= \frac{3}{5} \times \frac{24}{25} - \frac{7}{25} \times \frac{4}{5}$$

$$= \frac{44}{125}$$

1 correct formula with vaues

1 Answer

Some students got the formula wrong.

(d) Solve for $0 \leq \theta \leq 2\pi$:

(i) $\sin 2\theta = \frac{-1}{2}$

2

If $0 \leq \theta \leq 2\pi$ then $0 \leq 2\theta \leq 4\pi$.

$$2\theta = \sin^{-1}\left(\frac{-1}{2}\right)$$

(in quadrants 3,4 – with base angle $\sin^{-1}\left(\frac{1}{2}\right)$
$$= \frac{\pi}{6}$$

$$2\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}, \frac{23\pi}{6}$$

$$\theta = \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$$

(ii) $\sin^2 x - \sin x \cos x = 0$

$$\sin x(\sin x - \cos x) = 0$$

Hence $\sin x = 0$ or $\sin x - \cos x = 0$

$$x = 0, \pi, 2\pi$$

$$\sin x = \cos x$$

$$\tan x = 1 (\Rightarrow Q1, 3)$$

$$x = \frac{\pi}{4}, \frac{5\pi}{4}$$

Thus $x = 0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}, 2\pi$

(e)

(i) Write $\sqrt{3}\cos x - \sin x$ in form $r\cos(x + \alpha)$ where $0 < \alpha < \frac{\pi}{2}$.

2

$$r = \sqrt{\sqrt{3}^2 + 1^2} \text{ and } \tan \alpha = \frac{1}{\sqrt{3}}$$

$$= \sqrt{4}$$

$$= 2$$

$$\alpha = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{\pi}{6}$$

Hence $\sqrt{3}\cos x - \sin x = 2\cos\left(x + \frac{\pi}{6}\right)$

(ii) Hence or otherwise, solve $\sqrt{3}\cos x = 1 + \sin x$ for $0 < x < 2\pi$.

2

$$\sqrt{3}\cos x = 1 + \sin x$$

$$\sqrt{3}\cos x - \sin x = 1$$

$$2\cos\left(x + \frac{\pi}{6}\right) = 1 \quad \text{using part (i).}$$

For $0 < x < 2\pi$ then $\frac{\pi}{6} < x + \frac{\pi}{6} < \frac{13\pi}{6}$

① Base angle
and quadrants
resolved① correct
answersAbout a quarter of
the candidates
only got 2 of the 4
answers.

① factoring

① for cases
correct

① Answers

Factorizing a
problem for some:
 $\sin x(1 - \cos x)$
was a common
error.Many got the 3
answers for sin
but then forgot the
other two answers.① correct
 r & θ values

Mostly well done.

① correct
expressionSome students had
 $\left(x - \frac{\pi}{6}\right)$ ① correct base angle
and quadrants

$$\cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2} \text{ (i.e. Q1 and Q4)}$$

$$x + \frac{\pi}{6} = \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{3}, \frac{5\pi}{3}$$

$$x = \frac{\pi}{3} - \frac{\pi}{6}, \frac{5\pi}{3} - \frac{\pi}{6}$$

$$= \frac{\pi}{6}, \frac{3\pi}{2}$$

1 correct values

Mostly well done.

Question Four (Start a new booklet)

marks

(a) Use the rules of differentiation to differentiate with respect to x :

(i) $y = (3x + 4)(x^2 - 2x)$

2

$$y = 3x^3 - 2x^2 - 8x$$

$$y' = 9x^2 - 4x - 8$$

1 correct
expansion or
rule
1 correct
differentiation

Some who used
the product rule
also made silly
errors

Alternative Method (Product Rule):

$$y' = u \cdot dv + v \cdot du$$

where $u = 3x + 4$ $v = x^2 - 2x$

$$= (3x + 4)(2x - 2) + 3(x^2 - 2x) \quad du = 3 \quad dv = 2x - 2$$

$$= 6x^2 - 6x + 8x - 8 + 3x^2 - 6x$$

$$= 9x^2 - 4x - 8$$

(ii) $y = \sqrt{25 - x^2}$ (Answer as a surd)

2

$$y = (25 - x^2)^{\frac{1}{2}} \quad \text{where } u = 25 - x^2$$

$$y' = \frac{1}{2}(25 - x^2)^{-\frac{1}{2}} \times (-2x) \quad du = -2x$$

$$= \frac{-x}{\sqrt{25 - x^2}}$$

1 correct
Chain Rule
1 correct
differentiation

Some left the
negative sign out.

(iii) $y = \frac{x+1}{x^2+1}$

3

$$y' = \frac{v \cdot du - u \cdot dv}{v^2} \quad \text{where } u = x + 1 \quad v = x^2 + 1$$

$$du = 1 \quad dv = 2x$$

$$v^2 = (x^2 + 1)^2$$

$$y' = \frac{(x^2 + 1) \cdot 1 - (x + 1) \cdot 2x}{(x^2 + 1)^2}$$

$$= \frac{x^2 + 1 - (2x^2 + 2x)}{(x^2 + 1)^2}$$

$$= \frac{1 - 2x - 2x^2}{(x^2 + 1)^2}$$

1 correct
Quotient Rule
(or equivalent)

1 correct
algebra

1 correct
differentiation

Students need to
state the rule then
apply u and v in
the correct order.

(b) Differentiate $f(x) = (x + 1)^3$ from first principles.

3

The standard method:

$$\begin{aligned}
f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{((x+h)+1)^3 - (x+1)^3}{h} \\
&= \lim_{h \rightarrow 0} \frac{(x+(h+1))^3 - (x+1)^3}{h} \text{ ❶} \\
&= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2(h+1) + 3x(h+1)^2 + (h+1)^3 - (x^3 + 3x^2 + 3x + 1)}{h} \\
&= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3x^2 + 3xh^2 + 6xh + 3x + h^3 + 3h^2 + 3h + 1 - (x^3 + 3x^2 + 3x + 1)}{h} \text{ ❷} \\
&= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + 6xh + h^3 + 3h^2 + 3h}{h} \\
&= \lim_{h \rightarrow 0} 3x^2 + 3xh + 6x + h^2 + 3h + 3 \text{ ❸} \\
&= 3x^2 + 6x + 3
\end{aligned}$$

(marks for reaching steps indicated above)

Alternative methods:

Some students did realise other (simpler) methods, such as Difference of Two Cubes or Substitution:

Difference of Two Cubes:

$$\begin{aligned}
f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{(x+h+1)^3 - (x+1)^3}{h} \text{ ❶} \\
&= \lim_{h \rightarrow 0} \frac{(x+h+1-x-1)((x+h+1)^2 + (x+h+1)(x+1) + (x+1)^2)}{h} \\
&= \lim_{h \rightarrow 0} \frac{h((x+h+1)^2 + (x+h+1)(x+1) + (x+1)^2)}{h} \text{ ❷} \\
&= \lim_{h \rightarrow 0} ((x+h+1)^2 + (x+h+1)(x+1) + (x+1)^2) \\
&= (x+1)^2 + (x+1)(x+1) + (x+1)^2 \\
&= 3(x+1)^2 \text{ ❸}
\end{aligned}$$

Substitution:

$$u = x + 1$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(u+h)^3 - (u)^3}{h} \text{ ①}$$

$$= \lim_{h \rightarrow 0} \frac{u^3 + 3u^2h + 3uh^2 + h^3 - u^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3u^2h + 3uh^2 + h^3}{h} \text{ ①}$$

$$= \lim_{h \rightarrow 0} 3u^2 + 3uh + h^2$$

$$= 3u^2 \text{ ①}$$

$$= 3(x+1)^2$$

Generally (regardless of method used) the notation and setting out was terrible.

Many need to practice cubic expansions.

Some “fudged” their working to make the algebra come out correctly.

- (c) A block is cast into the shape of a right cylinder with a total surface area of $20\pi \text{ cm}^2$. The radius of the base is r cm and the height is h cm. The total surface area of a cylinder is given by $A = 2\pi r^2 + 2\pi rh$.

- (i) Express h in terms of r .

1



$$A = 2\pi r^2 + 2\pi rh$$

$$20\pi = 2\pi r^2 + 2\pi rh$$

$$10 - r^2 = rh$$

$$h = \frac{10}{r} - r$$

- (ii) Express the volume V in terms of r .

1

$$V = \pi r^2 h$$

$$= \pi r^2 \left(\frac{10}{r} - r \right)$$

$$= 10\pi r - \pi r^3$$

① setup and algebra correct

Generally well done.

- (iii) Find the value of r for which the volume is the greatest. Answer in exact form.

3

$$V = 10\pi r - \pi r^3$$

$$\frac{dV}{dr} = 10\pi - 3\pi r^2$$

$$\frac{d^2V}{dr^2} = -6\pi r$$

① Answer

Some carry on errors.

Stat pts when $\frac{dV}{dr} = 0$

$$0 = 10\pi - 3\pi r^2$$

$$3r^2 = 10$$

$$r = \pm \sqrt{\frac{10}{3}}$$

As $r > 0$, $\frac{d^2V}{dr^2} < 0 \Rightarrow \text{c.c.d. or max t.p.}$

Hence $r = \frac{\sqrt{30}}{3}$ for greatest volume.

❶ condition for stat pt correct

❶ r value justified

❶ r correct

Must state for stat. pts.

$$\frac{dV}{dr} = 0$$

Should state + and – answers and why –ve rejected.

Must test first or second derivative for min/max.

Need to conclude and actually answer the question posed.

Question Five (Start a new booklet)

marks

3(a) Find the general solution to $\cos 2\theta = 2 + \cos \theta$.

$$2\cos^2 \theta - 1 = 2 + \cos \theta$$

$$0 = 2\cos^2 \theta - \cos \theta - 3$$

$$= 2\cos^2 \theta - 3\cos \theta + 2\cos \theta - 3$$

$$= 2\cos \theta (2\cos \theta - 3) + (2\cos \theta - 3)$$

$$= (\cos \theta + 1)(2\cos \theta - 3)$$

Hence

$$\cos \theta + 1 = 0$$

$$\cos \theta = -1$$

$$\theta = 2n\pi \pm \cos^{-1}(-1)$$

$$= (2n \pm 1)\pi$$

$$2\cos \theta - 3 = 0$$

$$\cos \theta = \frac{3}{2}$$

no soln

① quadratic in $\cos$

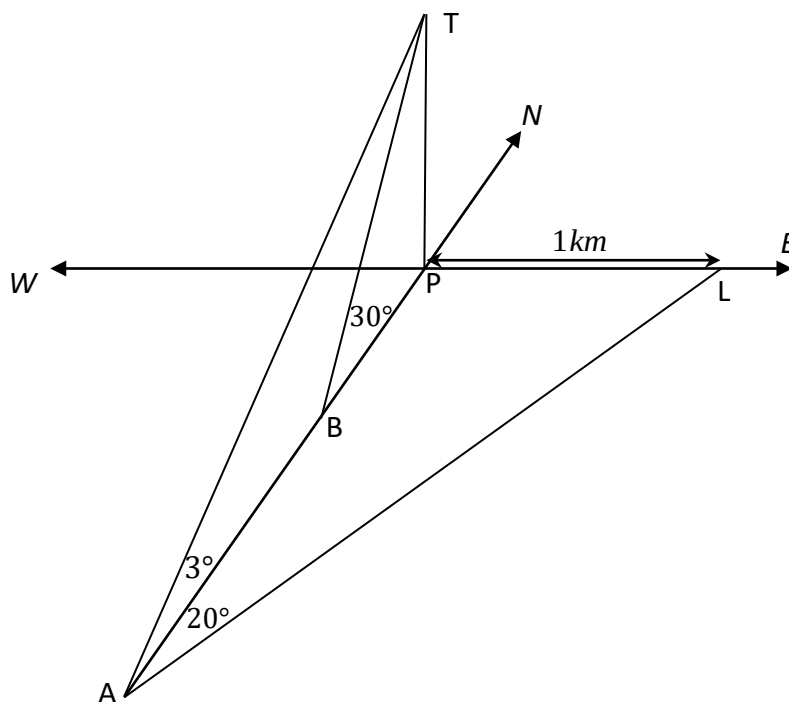
① cases resolved

① general soln correct

Generally well done.

(b) A boat is sailing due north from a point A towards a point P on the shore line. The shore line runs from west to east.

In the diagram, T represents a tree on a cliff vertically above P, and L represents a landmark on the shore. The distance PL is 1 km.

From A the point L is on a bearing of 020° , and the angle of elevation to T is 3° . After sailing for some time the boat reaches a point B, from which the angle of elevation to T is 30° .(i) Show that $BP = \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}$ **3**

$$\text{In } \triangle ALP: \tan 20^\circ = \frac{1}{AP}$$

$$AP = \cot 20^\circ$$

① correct AP with $\triangle$ identified

This is a SHOW question – the result for BP cannot be used in the working!

$$\text{In } \triangle ATP: \tan 3^\circ = \frac{PT}{AP}$$

$$PT = AP \tan 3^\circ$$

$$= \frac{\tan 3^\circ}{\tan 20^\circ}$$

$$\text{In } \triangle BTP: \tan 30^\circ = \frac{PT}{BP}$$

$$BP = \frac{PT}{\sqrt{3}}$$

$$= \sqrt{3}PT$$

$$= \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}$$

① correct
correct PT
with Δ
identified

AP, PT and BP
MUST be derived
from the given
information!
All working had
to be shown for
these derivations.

① correct
substitution
and resolution
of algebra

Many students
lacked
organisation in
their solutions.
Also, many quite
illegible.

This MUST
improve!

(ii) Find the exact distance AB .

$$AB = AP - BP$$

$$= \frac{1}{\tan 20^\circ} - \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}$$

$$= \frac{1 - \sqrt{3} \tan 3^\circ}{\tan 20^\circ}$$

① answer (in exact
form)

Many lost the mark
because their answer was
not in EXACT FORM as
specified by the
question!

1

(c) If $t = \tan\left(\frac{\theta}{2}\right)$ solve for t the equation $12 \tan \theta = 5$ for $\pi \leq \theta < \frac{3\pi}{2}$

4

$$12 \times \frac{2t}{1-t^2} = 5$$

$$24t = 5 - 5t^2$$

$$0 = 5t^2 + 24t - 5$$

$$= 5t^2 + 25t - t - 5$$

$$= 5t(t+5) - (t+5)$$

$$= (t+5)(5t-1)$$

$$\text{Hence } t+5=0 \text{ or } 5t-1=0$$

$$t = -5 \quad 5t = 1$$

$$t = \frac{1}{5}$$

$$\text{With } \pi \leq \theta < \frac{3\pi}{2}, \text{ then } \frac{\pi}{2} \leq \frac{\theta}{2} < \frac{3\pi}{4} \text{ (or}$$

$$90^\circ \leq \frac{\theta}{2} < 135^\circ) \text{ leads to:}$$

$$\frac{\theta}{2} = \tan^{-1}(-5) \quad \frac{\theta}{2} = \tan^{-1}\left(\frac{1}{5}\right)$$

$$\approx 101^\circ$$

$$\approx 11^\circ$$

① quadratic
in t

① two results
correct

① testing
domain

Most students
obtained the first
two marks easily.
Some need to
learn the 't'
results!
For full marks,
students needed to
refer to the
domain in their
answer.

Most students lost
a mark for NOT
ANSWERING the
question! i.e. to
solve for 't'
values, NOT θ
values!

Hence $t = -5$ (reject $t = \frac{1}{5}$ as out of the required domain)

① correct t value

(d) Sketch $y = x + \frac{4}{x-1}$, showing asymptotes, intercepts and stationary points having determined their nature.

4

$$y - x = \frac{4}{x-1}$$

$$(y - x)(x - 1) = 4 \Rightarrow y \neq x, x \neq 1$$

Hence asymptotes are $y = x$ and $x = 1$.

$$\frac{dy}{dx} = 1 - \frac{4}{(x-1)^2}$$

$$= \frac{(x-1)^2 - 4}{(x-1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{8}{(x-1)^3}$$

$$\frac{dy}{dx} = 0 \Rightarrow$$

$$0 = (x-1)^2 - 4$$

$$4 = (x-1)^2$$

$$\pm 2 = x - 1$$

$$x = 1 \pm 2$$

$$= -1, 3$$

$$x = -1:$$

$$y = -3$$

$$\frac{d^2y}{dx^2} = \frac{8}{(-1-1)^3}$$

$$< 0 \Rightarrow ccd$$

$$\therefore (-1, -3) \text{ a max t.p.}$$

$$x = 3:$$

$$y = 5$$

$$\frac{d^2y}{dx^2} = \frac{8}{(3-1)^3}$$

$$> 0 \Rightarrow ccu$$

$$\therefore (3, 5) \text{ a min t.p.}$$

$$\text{Also } x = 0 \Rightarrow y = -4$$

① asymptotes correctly identified

① Stat pts correct

① nature of stat pts correctly done

$y \neq x, x \neq 1$
are NOT asymptotes.

Most students were able to find the correct stationary points. However, testing for nature of stationary points needs to be done correctly – make sure you are using the sign of y' or y'' to determine the nature.

Graph:
Overall, poorly done. In many cases:

- i) The curve did not approach the asymptotes
- ii) No relative scale was shown
- iii) No y-intercept shown
- iv) Asymptotes were drawn solid, not dotted
- v) The graph size was too small and/or cramped and unclear.
- vi) Some graphs looked like a string of spaghetti dropped down on the page in roughly the right shape!

Take more care with graphing!

❶ diagram (including y intercept) correct